PHY175: PHYSICS FOR COMPUTER SCIENCE

Midterm Master 200-MCQ Question Bank & High-Yield Revision Guide

Units 1, 2 & 3 | 1st Year CSE Core | Target: 30 / 30 Marks

■ 3-Hour Rapid Mastery Schedule & Exam Blueprint

Time Window	Unit / Module	Core Focus Topics	Target Weightage
00:00 - 01:00	**Unit 1: Solid State Physics**	Drude-Lorentz, $\sigma=(ne^2\tau)/m$, Wiedemann-Franz, Fermi Level, Mass Action, Hall Effect	**10 MCQs (10 M)**
01:00 - 02:00	**Unit 2: Devices & Optics**	KCL/KVL, Diode, Zener vs Avalanche, BJT (α,β,γ), CMOS Inverter, Solar Cell, Fiber Optics (NA, V-no), CPU vs GPU	**10 MCQs (10 M)**
02:00 - 03:00	**Unit 3: Digital Electronics**	Number Systems, 1's & 2's Comp, Excess-3, Gray Code, Universal Gates, De Morgan, K-Map quads/corners	**10 MCQs (10 M)**
03:00 - 03:45	**Mock Exam & Drill**	Solve 200 MCQs drill, test tricky trap questions, memorize formula cheat sheet	**30 / 30 MARKS**

■ Key Formulas & Traps Cheat Sheet

Unit	Concept / Quantity	Formula / Mathematical Relation	Key Insight / Exam Trap
Unit 1	Conductivity & Mobility	$\sigma = (n e^2 \tau) / m = n e \mu$ $v_d = \mu E = (e E \tau) / m$	Classical theory failed: predicted $\rho \propto \sqrt{T}$ instead of $\rho \propto T$
Unit 1	Wiedemann-Franz Law	$K / (\sigma T) = L = 2.44 \times 10$■■ $W \cdot \Omega / K^2$	L is constant for all metals at given T
Unit 1	Einstein Relation	$D / \mu = (k_B T) / q = V_T \approx 26$ mV	At 300 K, thermal voltage $V_T \approx 0.026$ V
Unit 1	Mass Action & Fermi Level	$n \cdot p = n_i^2$ $f(E) = 1 / [1 + \exp((E-E_F)/kBT)]$	At T > 0K, f(EF) = 0.5 (50% probability)
Unit 1	Hall Effect	$V_H = (R_H B I) / t$ $R_H = -1/(ne)$ [N-type], $+1/(pe)$ [P-type] $\mu_H = \sigma \lvert R_H \rvert$	$R_H < 0$ proves electrons are majority carriers
Unit 2	Diode & Breakdown	$V_{barrier}$: Si = 0.7V, Ge = 0.3V Zener: <6V (neg temp coeff) Avalanche: >6V (pos temp coeff)	I_0 doubles every 10°C rise in temp
Unit 2	BJT Current Gains	$I_E = I_B + I_C$ $\beta = \alpha / (1 - \alpha)$; $\alpha = \beta / (1 + \beta)$ $\gamma = 1 + \beta$	Doping: E > C > B Size: C > E > B
Unit 2	Solar Cell	$FF = (V_{mp} I_{mp}) / (V_{oc} I_{sc})$ $\eta = (V_{oc} I_{sc} FF / P_{in}) \times 100\%$	Operates in 4th quadrant (power generator)
Unit 2	Optical Fiber	$NA = \sqrt{(n_1^2 - n_2^2)} \approx n_1\sqrt{(2\Delta)}$ $V = (2\pi a / \lambda) NA$ Modes: $M \approx V^2/2$ (Step), $V^2/4$ (GRIN)	Single mode cutoff: $V \leq 2.405$ Eliminates intermodal dispersion
Unit 3	2's Complement	2's comp = 1's comp + 1 Range (8-bit): -128 to +127	Unique zero; End carry discarded during subtraction
Unit 3	Universal Gates	XOR: 4 NANDs or 5 NORs XNOR: 5 NANDs or 4 NORs AND: 2 NANDs / 3 NORs; OR: 3 NANDs / 2 NORs	Excess-3 is self-complementing Gray code is unit distance
Unit 3	Boolean & K-Maps	A + A'B = A + B AB + A'C + BC = AB + A'C (Consensus) Corner Quad (0,2,8,10) = B'D'	Don't cares (d) used only to enlarge groups

■ Complete 200-MCQ Master Question Bank (Units 1, 2 & 3)

Each question includes all 4 options, the verified correct key, and detailed pedagogical rationale.

■ UNIT 1: SOLID STATE PHYSICS

Q1. According to the Classical Free Electron Theory of Drude and Lorentz (1900), valence electrons in metals are assumed to behave as:

A) Quantum bound harmonic oscillators

B) A classical ideal gas of free particles colliding elastically with positive ion cores

C) Immobile particles tightly locked to lattice sites

D) Fermions strictly obeying the Pauli exclusion principle at all temperatures

Correct Answer: Option B | *Drude-Lorentz theory treats conduction electrons as a classical gas of free particles undergoing elastic collisions with stationary positive ion cores.*

Q2. The microscopic form of Ohm's Law relates current density J and electric field E through electrical conductivity σ as:

A) $J = \sigma / E$

B) $J = \sigma \cdot E$

C) $E = \sigma \cdot J$

D) $J = \sigma^2 \cdot E$

Correct Answer: Option B | *Microscopic Ohm's Law is $J = \sigma E$, derived from $J = n \cdot e \cdot vd$ and $vd = (e \cdot E \cdot \tau / m)$.*

Q3. The electrical conductivity σ of a metal in classical free electron theory is given by:

A) $\sigma = (n e^2 \tau) / m$

B) $\sigma = (n e \tau) / m^2$

C) $\sigma = (n^2 e \tau) / m$

D) $\sigma = m / (n e^2 \tau)$

Correct Answer: Option A | *From $J = n \cdot e \cdot vd = n \cdot e \cdot (e \cdot E \cdot \tau / m) = (n \cdot e^2 \cdot \tau / m) \cdot E$, comparing with $J = \sigma \cdot E$ gives $\sigma = (n \cdot e^2 \cdot \tau) / m$.*

Q4. The Wiedemann-Franz Law states that for metals at a given temperature, the ratio of thermal conductivity (K) to electrical conductivity (σ) is:

A) Inversely proportional to absolute temperature T

B) Directly proportional to absolute temperature T ($K/\sigma \propto T$)

C) Proportional to T^2

D) Independent of temperature

Correct Answer: Option B | *Wiedemann-Franz Law: $K / (\sigma T) = L$ (Lorentz Number = 2.44×10■■ $W \cdot \Omega / K^2$), meaning K/σ is directly proportional to T.*

Q5. Which of the following experimental observations represents a critical failure of classical Drude-Lorentz theory?

A) Inability to derive Ohm's law

B) Predicting an electronic molar heat capacity of (3/2)R, whereas the experimental value is ~0.01R at 300 K

C) Explanation of high electrical conductivity in copper

D) Explaining why metals reflect light

Correct Answer: Option B | *Classical equipartition predicted $Cv,el = 3/2\ R$, but experimentally it is ~100 times smaller because only a tiny fraction of electrons near the Fermi level absorb thermal energy.*

Q6. Classical free electron theory predicts the temperature dependence of electrical resistivity of metals as $\rho \propto \sqrt{T}$, whereas experiments show:

A) $\rho \propto 1/T$
B) $\rho \propto T$
C) $\rho \propto T^3$
D) ρ is independent of T

Correct Answer: Option B | *Experimentally, resistivity of pure metals increases linearly with temperature ($\rho \propto T$) due to lattice vibration (phonon) scattering.*

Q7. Quantum Free Electron Theory was proposed by Arnold Sommerfeld in 1928 by applying:

A) Maxwell-Boltzmann classical statistics
B) Fermi-Dirac quantum statistics and Pauli Exclusion Principle
C) Bohr's planetary model of atom
D) Newton's laws of motion exclusively

Correct Answer: Option B | *Sommerfeld resolved classical discrepancies by treating electrons as quantum fermions obeying Pauli exclusion and Fermi-Dirac statistics.*

Q8. The drift velocity vd of free electrons in a conductor is mathematically defined as:

A) $vd = \mu / E$
B) $vd = \mu \cdot E = (e E \tau / m)$
C) $vd = e E / (m \tau)$
D) $vd = m / (e E \tau)$

Correct Answer: Option B | *Drift velocity is the net velocity acquired in an electric field: $vd = \mu E = (e \tau / m) E$.*

Q9. Relaxation time (τ) of a conduction electron is defined as the average time:

A) Taken by an electron to traverse the conductor length
B) Elapsed between two successive collisions with lattice ions
C) Required for an electron to escape into vacuum
D) Required for an electron to recombine with a hole

Correct Answer: Option B | *Relaxation time τ is the mean time interval between two successive collisions of a conduction electron.*

Q10. Mobility (μ) of charge carriers is defined as:

A) Drift velocity per unit electric field (vd / E)
B) Electric field per unit drift velocity (E / vd)
C) Current density per unit charge (J / q)
D) Diffusion coefficient per unit time (D / t)

Correct Answer: Option A | *Mobility is defined as drift velocity acquired per unit applied electric field: $\mu = vd / E = e \tau / m$.*

Q11. Drift current is produced by _____, whereas diffusion current is produced by _____.

A) Carrier concentration gradient; Applied electric field
B) Applied electric field; Carrier concentration gradient
C) Magnetic field; Thermal gradient
D) Mechanical stress; Optical illumination

Correct Answer: Option B | *Drift current arises from carrier acceleration in an electric field, while diffusion current arises from spatial variations in carrier concentration.*

Q12. The Einstein relation connecting diffusion coefficient (D) and mobility (μ) of charge carriers at temperature T is:

A) $D / \mu = kB T / q = VT$
B) $\mu / D = kB T / q$
C) $D \cdot \mu = kB T / q$
D) $D / \mu = q / (kB T)$

Correct Answer: Option A | *Einstein relation: $D / \mu = (kB T / q) = VT$ (Thermal voltage $\approx$ 26 mV at 300 K).*

Q13. The total current density J in a semiconductor containing both electrons and holes is given by:

A) $J = q(n \mu n + p \mu p)E + q Dn (dn/dx) - q Dp (dp/dx)$
B) $J = q(n \mu n - p \mu p)E + q Dn (dn/dx) + q Dp (dp/dx)$
C) $J = q(n \mu n + p \mu p)E - q Dn (dn/dx) - q Dp (dp/dx)$
D) $J = \sigma E + q Dn (dp/dx)$

Correct Answer: Option A | *Total current $J = J_{drift} + J_{diff} = q(n \mu n + p \mu p)E + q Dn (dn/dx) - q Dp (dp/dx)$. The minus sign on hole diffusion reflects positive charge moving down the gradient.*

Q14. The forbidden energy gap (Eg) in solids is defined as the energy difference between:

A) Bottom of Valence Band and Top of Conduction Band
B) Top of Valence Band (Ev) and Bottom of Conduction Band (Ec)
C) Fermi level and vacuum level
D) Donor level and acceptor level

Correct Answer: Option B | *Eg = Ec - Ev, the forbidden energy span between the highest valence state and lowest conduction state.*

Q15. At room temperature (300 K), the forbidden energy gaps of Silicon (Si) and Germanium (Ge) are approximately:

A) 0.67 eV and 1.1 eV
B) 1.1 eV and 0.67 eV
C) 1.42 eV and 5.4 eV
D) 0 eV and 1.1 eV

Correct Answer: Option B | *Eg(Si) ≈ 1.1 eV and Eg(Ge) ≈ 0.67 eV at room temperature.*

Q16. In conductors (metals), the valence band and conduction band:

A) Are separated by an energy gap greater than 5 eV
B) Overlap or the valence band is partially filled
C) Are completely empty at 0 K
D) Are separated by exactly 1.1 eV

Correct Answer: Option B | *In metals, the conduction and valence bands overlap (Eg = 0), providing abundant free electrons for conduction.*

Q17. Insulators like diamond have a forbidden energy gap Eg that is typically:

A) Exactly 0 eV
B) Around 1 eV
C) Greater than 3 to 5 eV (~5.4 eV for diamond)
D) Negative

Correct Answer: Option C | *Insulators have wide bandgaps (Eg > 3 to 5 eV; diamond is ~5.4 eV), preventing thermal excitation across the gap.*

Q18. The temperature coefficient of resistance (α) for intrinsic semiconductors is:

A) Positive (resistance increases with temperature)
B) Negative (resistance decreases / conductivity increases exponentially with temperature)
C) Zero at all temperatures
D) Infinite

Correct Answer: Option B | *Semiconductors have a negative temperature coefficient of resistance ($\alpha < 0$); thermal energy generates electron-hole pairs, drastically boosting conductivity.*

Q19. At absolute zero temperature (0 K), an ideal pure intrinsic semiconductor behaves as a:

A) Superconductor
B) Perfect conductor
C) Perfect insulator
D) Semiconductor with high current

Correct Answer: Option C | *At 0 K, no thermal energy exists to break covalent bonds; VB is completely full, CB is completely empty.*

Q20. An intrinsic semiconductor crystal (such as pure Silicon) is held together by:

A) Ionic bonds
B) Covalent bonds
C) Metallic bonds
D) Hydrogen bonds

Correct Answer: Option B | *Silicon and Germanium atoms (Group IV) share 4 valence electrons with 4 neighboring atoms via covalent bonds.*

Q21. In an intrinsic semiconductor under thermal equilibrium, the carrier concentrations satisfy:

A) $n > p$
B) $p > n$
C) $n = p = n_i$
D) $n \cdot p = 0$

Correct Answer: Option C | *Thermal bond rupture generates one electron in CB and one hole in VB simultaneously, so $n = p = n_i$.*

Q22. The Mass Action Law states that in any semiconductor under thermal equilibrium:

A) $n + p = n_i$
B) $n \cdot p = n_i^2$
C) $n / p = n_i^2$
D) $n \cdot p = 2 n_i$

Correct Answer: Option B | *Mass Action Law: $n \cdot p = n_i^2$ at a constant temperature, valid for both intrinsic and extrinsic semiconductors.*

Q23. Silicon is preferred over Germanium in modern semiconductor electronics primarily because:

A) Si has a smaller bandgap than Ge
B) Si has a larger bandgap (1.1 eV vs 0.67 eV), resulting in much lower reverse leakage current and higher temperature stability
C) Si has higher electron mobility than Ge
D) Ge is completely non-conductive

Correct Answer: Option B | *Si has larger Eg, smaller reverse saturation current, higher operating temperature (up to 200°C vs 85°C for Ge), and forms stable native SiO2.*

Q24. When pure Silicon is doped with a Pentavalent impurity (such as Phosphorus, Arsenic, or Antimony):

A) P-type semiconductor is created
B) N-type semiconductor is created with donor energy level Ed just below the conduction band
C) The crystal becomes negatively charged overall
D) Carrier concentration drops to zero

Correct Answer: Option B | *Pentavalent atoms donate free electrons to CB, creating an N-type semiconductor with donor level Ed ~0.05 eV below Ec.*

Q25. When pure Silicon is doped with a Trivalent impurity (such as Boron, Aluminum, Gallium, or Indium):

A) N-type semiconductor is formed
B) P-type semiconductor is formed with acceptor energy level Ea just above the valence band
C) The crystal becomes positively charged overall
D) Electron concentration exceeds hole concentration

Correct Answer: Option B | *Trivalent atoms accept electrons to complete bonds, creating holes in VB and forming P-type material with Ea ~0.05 eV above Ev.*

Q26. An N-type semiconductor as a whole is:

A) Negatively charged
B) Positively charged
C) Electrically neutral
D) Positively charged only at high temperatures

Correct Answer: Option C | *An N-type crystal is electrically neutral because fixed positive donor ions exactly balance free electrons.*

Q27. In an N-type semiconductor, the majority carriers are _____ and minority carriers are _____.

A) Holes; Electrons
B) Electrons; Holes
C) Protons; Neutrons
D) Ions; Holes

Correct Answer: Option B | *In N-type material, electrons are majority carriers ($n \approx N_D$) and holes are minority carriers ($p = n_i^2 / N_D$).*

Q28. The Fermi-Dirac distribution function f(E) gives the probability that an available energy state E is:

A) Empty at absolute zero
B) Occupied by an electron at temperature T
C) Generating a photon
D) Radiating thermal energy

Correct Answer: Option B | *$f(E) = 1 / [1 + \exp((E - E_F)/k_B T)]$ represents the probability of electron occupancy of state E at temperature T.*

Q29. At any temperature T > 0 K, the probability of finding an electron at the Fermi level E = EF is exactly:

A) 1.0 (100%)
B) 0.5 (50%)
C) 0.0 (0%)
D) 0.25 (25%)

Correct Answer: Option B | *$f(E_F) = 1 / [1 + \exp((E_F - E_F)/k_B T)] = 1 / [1 + \exp(0)] = 1 / 2 = 0.5$.*

Q30. In an intrinsic semiconductor, the Fermi level EF lies:

A) Near the conduction band edge Ec
B) Near the valence band edge Ev
C) Exactly at the center of the forbidden bandgap (Ec + Ev)/2
D) Inside the conduction band

Correct Answer: Option C | *Since electron and hole densities are equal in intrinsic semiconductor, EF = (Ec + Ev)/2.*

Q31. In an N-type semiconductor, as doping concentration increases, the Fermi level EF:

A) Shifts downward towards the valence band
B) Shifts upward closer to the conduction band edge Ec
C) Remains strictly at the middle of the bandgap
D) Drops into the valence band

Correct Answer: Option B | *Adding donor impurities increases electron density, pushing EF upward toward Ec.*

Q32. In a P-type semiconductor, the Fermi level EF lies:

A) Closer to the valence band edge Ev
B) Closer to the conduction band edge Ec
C) Outside the crystal
D) At the exact top of the conduction band

Correct Answer: Option A | *Adding acceptor impurities increases hole concentration, pulling EF down toward Ev.*

Q33. When a magnetic field B is applied perpendicular to a current I in a semiconductor, the resulting transverse electric field is called the:

A) Seebeck field
B) Hall electric field (EH)
C) Peltier field
D) Faraday field

Correct Answer: Option B | *The transverse field established across the conductor to balance the Lorentz force is the Hall electric field EH.*

Q34. The Hall voltage VH across a sample of thickness t carrying current I in magnetic field B is given by:

A) VH = (RH · B · I) / t
B) VH = (RH · t) / (B · I)
C) VH = RH · B · I · t
D) VH = (B · I) / (RH · t)

Correct Answer: Option A | *VH = EH · w = (RH · J · B) · w = RH (I / (w t)) B w = (RH B I) / t.*

Q35. The Hall coefficient RH for an N-type semiconductor is:

A) Positive (RH > 0)
B) Negative (RH = -1/(n e) < 0)
C) Exactly zero
D) Imaginary

Correct Answer: Option B | *For electrons (charge -e), RH = -1 / (n e), which is negative.*

Q36. The Hall coefficient RH for a P-type semiconductor is:

A) Positive (RH = +1/(p e) > 0)
B) Negative (RH < 0)
C) Zero
D) Dependent only on magnetic field

Correct Answer: Option A | *For holes (charge +e), RH = +1 / (p e), which is positive.*

Q37. Hall mobility μH of charge carriers is related to conductivity σ and Hall coefficient RH by:

A) μH = σ / |RH|
B) μH = σ · |RH|
C) μH = |RH| / σ
D) μH = σ² · |RH|

Correct Answer: Option B | *Since σ = n e μ and |RH| = 1/(n e), we have μH = σ · |RH|.*

Q38. A copper wire carries current I = 5 A. If electron density n = 8.5 × 10²■ m■³ and cross-sectional area A = 1.0 × 10■■ m², the electron drift speed vd is: (e = 1.6 × 10■¹■ C)

A) 3.68 × 10■■ m/s
B) 3.68 × 10■ m/s
C) 1.25 × 10■² m/s
D) 3.0 × 10■ m/s

Correct Answer: Option A | *vd = I / (n e A) = 5 / (8.5e28 * 1.6e-19 * 1e-6) = 5 / 13600 = 3.68 × 10■■ m/s.*

Q39. A Silicon sample is doped with ND = 10^{1}■ cm■^{3} of Phosphorus. If ni = 1.5 × 10^{1}■ cm■^{3}, the minority hole concentration p is:

A) 1.5 × 10^{3} cm■^{3}
B) 2.25 × 10^{3} cm■^{3}
C) 2.25 × 10^{1}■ cm■^{3}
D) 10^{1}■ cm■^{3}

Correct Answer: Option B | *p = ni² / ND = (1.5 × 10^{1}■)² / 10^{1}■ = 2.25 × 10^{2}■ / 10^{1}■ = 2.25 × 10^{3} cm■^{3}.*

Q40. A semiconductor slab of thickness 0.5 mm carrying 10 mA in B = 0.5 T has RH = 0.05 m³/C. The Hall voltage VH is:

A) 0.5 mV
B) 5.0 mV
C) 0.05 V
D) 0.5 V

Correct Answer: Option A | *VH = (RH B I) / t = (0.05 * 0.5 * 0.01) / (0.5 × 10■^{3}) = 0.00025 / 0.0005 = 0.5 mV.*

Q41. If the Hall coefficient RH of a semiconductor is +1.25 × 10■■ m³/C, the carrier concentration is:

A) 5 × 10^{22} m■^{3} holes
B) 5 × 10^{22} m■^{3} electrons
C) 1.25 × 10^{2}■ m■^{3} holes
D) 8 × 10^{21} m■^{3} electrons

Correct Answer: Option A | *Positive RH means holes (P-type). p = 1 / (e RH) = 1 / (1.6e-19 * 1.25e-4) = 5 × 10^{22} m■^{3}.*

Q42. Which of the following is NOT an application of the Hall effect?

A) Determining whether a semiconductor is N-type or P-type
B) Measuring carrier concentration and mobility
C) Measuring unknown magnetic field B
D) Directly measuring the speed of light in vacuum

Correct Answer: Option D | *Hall effect measures carrier polarity, concentration, mobility, and magnetic fields, not the speed of light.*

Q43. In a pure semiconductor, when temperature increases from 300 K to 400 K, the electrical conductivity:

A) Decreases linearly
B) Increases exponentially
C) Drops to zero
D) Remains strictly constant

Correct Answer: Option B | *Conductivity increases exponentially as σ ∝ exp(-Eg / 2kB T) due to rapid generation of electron-hole pairs.*

Q44. In an N-type semiconductor, the donor energy level Ed lies:

A) In the middle of the forbidden gap
B) Just below the bottom of the conduction band Ec
C) Just above the top of the valence band Ev
D) Inside the valence band

Correct Answer: Option B | *Donor level Ed is very close to Ec (~0.05 eV in Si), making ionization very easy at room temperature.*

Q45. In a P-type semiconductor, the acceptor energy level Ea lies:

A) In the middle of the conduction band
B) Just above the top of the valence band Ev
C) Just below the bottom of the conduction band Ec
D) Outside the energy band diagram

Correct Answer: Option B | *Acceptor level Ea lies just above Ev (~0.05 eV in Si), readily accepting electrons from VB.*

Q46. The unit of Hall coefficient RH in SI units is:

A) m³ / C
B) V / m
C) Ω · m
D) C / m²

Correct Answer: Option A | *RH = 1 / (n q) = 1 / (m■^{3} · C) = m³ / C (or m³ / A·s).*

Q47. The unit of carrier mobility μ is:

A) m² / (V · s)
B) V / (m · s)
C) m / s²
D) Ω · m²

Correct Answer: Option A | *μ = vd / E = (m/s) / (V/m) = m² / (V·s) (or cm² / V·s).*

Q48. Thermal voltage VT at room temperature (T = 300 K) is approximately equal to:

A) 0.7 V
B) 0.3 V
C) 26 mV (0.0259 V)
D) 1.1 V

Correct Answer: Option C | *VT = kB T / q = (1.38 × 10■²³ * 300) / (1.6 × 10■¹■) ≈ 0.02586 V ≈ 26 mV.*

Q49. When a semiconductor is illuminated by light with photon energy hν < Eg:

A) Intense electron-hole pairs are created
B) No electron-hole pairs are generated (semiconductor is transparent)
C) The semiconductor becomes a superconductor
D) The bandgap collapses to zero

Correct Answer: Option B | *Photons with energy below Eg cannot excite electrons across the bandgap, so the material is transparent.*

Q50. At very high temperatures, an extrinsic doped semiconductor behaves as:

A) An insulator
B) An intrinsic semiconductor (intrinsic carrier generation overwhelms dopant density)
C) A superconductor
D) A pure metal with overlapping bands

Correct Answer: Option B | *At high T, thermally generated ni >> ND or NA, causing the material to behave like an intrinsic semiconductor.*

Q51. Mean free path λ of conduction electrons is related to drift velocity vd and relaxation time τ as:

A) $\lambda = v_d \cdot \tau$
B) $\lambda = v_d / \tau$
C) $\lambda = \tau / v_d$
D) $\lambda = v_d^2 \cdot \tau$

Correct Answer: Option A | *Mean free path λ is the average distance between collisions: λ = vd · τ.*

Q52. For an energy state E located 3 kB T above the Fermi level (E - EF = 3 kB T), the Fermi-Dirac occupation probability f(E) is:

A) 1 / (1 + e³)
B) 1 / (1 + e■³)
C) 0.5
D) 1.0

Correct Answer: Option A | *f(E) = 1 / [1 + exp((E - EF)/kB T)] = 1 / [1 + exp(3)] ≈ 1 / (1 + 20.08) ≈ 0.047 (4.7%).*

Q53. For an energy state E located 3 kB T below the Fermi level (E - EF = -3 kB T), the probability of finding a hole (unoccupied state) is:

A) 1 / (1 + e³)
B) 1 / (1 + e■³)
C) 0.5
D) 0.95

Correct Answer: Option A | *Probability of being empty 1 - f(E) = 1 - 1/(1 + e■³) = e■³ / (1 + e■³) = 1 / (1 + e³).*

Q54. Which of the following elements is a trivalent acceptor dopant for Silicon?

A) Phosphorus (P)
B) Arsenic (As)
C) Boron (B)
D) Antimony (Sb)

Correct Answer: Option C | *Boron (Group III) is trivalent; P, As, Sb (Group V) are pentavalent donors.*

Q55. Which of the following elements is a pentavalent donor dopant for Silicon?

A) Boron (B)
B) Gallium (Ga)
C) Arsenic (As)
D) Indium (In)

Correct Answer: Option C | *Arsenic (Group V) is pentavalent; B, Ga, In (Group III) are trivalent.*

Q56. In a Hall effect setup, if current I flows in +x direction and magnetic field B is applied in +z direction, electrons experience Lorentz force in:

A) +y direction
B) -y direction
C) +z direction
D) -x direction

Correct Answer: Option B | *Electrons drift in -x direction. Lorentz force F = -e (v × B) = -e ((-v x■) × (B z■)) = -e (v B y■) = -y direction.*

Q57. If electrical conductivity of a semiconductor is σ = 100 S/m and RH = 0.02 m³/C, the carrier mobility is:

A) 2 m²/(V·s)
B) 5000 m²/(V·s)
C) 0.0002 m²/(V·s)
D) 200 m²/(V·s)

Correct Answer: Option A | *μ = σ · RH = 100 * 0.02 = 2 m² / (V·s).*

Q58. The concept of 'effective mass' of an electron arises in:

A) Drude classical theory
B) Band theory of solids due to electron interaction with periodic crystal lattice potential
C) Ideal gas laws
D) Coulomb electrostatic law

Correct Answer: Option B | *Effective mass m* accounts for the forces exerted on the electron by the periodic crystal potential.*

Q59. In Germanium (Ge), the donor level Ed lies approximately _____ below the conduction band edge Ec.

A) 0.01 eV
B) 0.05 eV
C) 0.67 eV
D) 1.1 eV

Correct Answer: Option A | *Ed in Ge is ~0.01 eV below Ec (compared to ~0.05 eV in Si).*

Q60. Direct bandgap semiconductors (like GaAs) differ from indirect bandgap semiconductors (like Si) because:

A) Conduction band minimum and Valence band maximum occur at the same crystal momentum (k = 0), allowing efficient photon emission
B) They have zero bandgap
C) They cannot be doped
D) Conduction band is always full

Correct Answer: Option A | *Direct bandgap semiconductors allow direct radiative electron-hole recombination without phonon assistance, making them ideal for LEDs/lasers.*

Q61. The Fermi energy EF of a metal at T = 0 K is:

A) The energy of the lowest conduction state
B) The highest occupied energy state at 0 K
C) Zero for all metals
D) Infinite

Correct Answer: Option B | *At 0 K, all energy states up to EF are 100% occupied, and all states above EF are completely empty.*

Q62. A hole in semiconductor physics is treated as:

A) A positively charged quasiparticle with charge +e and positive effective mass
B) A real free proton in the nucleus
C) An electron moving with negative velocity
D) A neutral neutron

Correct Answer: Option A | *A hole is an empty electron state in the valence band behaving as a mobile positive quasiparticle with charge +e.*

Q63. If temperature of a pure semiconductor is increased, the Fermi level EF:

A) Stays exactly at the center of the bandgap
B) Deviates slightly depending on effective masses of electrons and holes: EF = (Ec+Ev)/2 + (3/4) kB T ln(mh*/me*)
C) Jumps immediately to conduction band
D) Becomes zero

Correct Answer: Option B | *EF = (Ec+Ev)/2 + (3/4) kB T ln(mh*/me*); if mh* ≈ me*, it stays essentially at the exact center.*

Q64. In a metal, electrical resistivity ρ is proportional to relaxation time τ as:

A) $\rho \propto \tau$
B) $\rho \propto 1 / \tau$
C) $\rho \propto \tau^2$
D) ρ is independent of τ

Correct Answer: Option B | *Since $\sigma = n e^2 \tau / m$, resistivity $\rho = 1/\sigma = m / (n e^2 \tau) \propto 1/\tau$.*

Q65. If a sample has a negative Hall coefficient $R_H < 0$, the sign of majority charge carriers is:

A) Positive (+q)
B) Negative (-q, electrons)
C) Neutral
D) Indeterminate

Correct Answer: Option B | *$R_H < 0$ definitively proves that the majority charge carriers are electrons.*

■ UNIT 2: ELECTRICAL LAWS, DEVICES, OPTICS & COMPUTING

Q66. Kirchhoff's Current Law (KCL) is based on which fundamental conservation law?

A) Conservation of Energy
B) Conservation of Electric Charge ($dq/dt = 0$ at node)
C) Conservation of Linear Momentum
D) Conservation of Angular Momentum

Correct Answer: Option B | *KCL states that algebraic sum of currents entering a node is zero ($\sum I = 0$), representing conservation of charge.*

Q67. Kirchhoff's Voltage Law (KVL) is based on which fundamental conservation law?

A) Conservation of Charge
B) Conservation of Energy (work done moving around closed loop is zero)
C) Conservation of Mass
D) Newton's Third Law

Correct Answer: Option B | *KVL states that algebraic sum of voltages around any closed loop is zero ($\sum V = 0$), representing conservation of energy.*

Q68. In an unbiased PN junction at thermal equilibrium, the depletion region contains:

A) Abundant free electrons and holes
B) Immobile ionized donor (+q) and acceptor (-q) ions devoid of free mobile carriers
C) Neutral Silicon atoms only
D) Pure metallic contacts

Correct Answer: Option B | *Diffusion leaves uncompensated immobile positive donor ions on N-side and negative acceptor ions on P-side.*

Q69. The barrier potential (built-in potential V_0) at 300 K is approximately _____ for Silicon and _____ for Germanium.

A) 0.3 V and 0.7 V
B) 0.7 V and 0.3 V
C) 1.1 V and 0.67 V
D) 2.0 V and 1.5 V

Correct Answer: Option B | *Barrier potential is ~0.7 V for Silicon and ~0.3 V for Germanium at room temperature.*

Q70. When a PN junction is forward biased (P connected to positive, N to negative):

A) Depletion width widens and barrier height increases
B) Depletion width narrows, barrier potential reduces (V_0 - V_F), and diffusion current increases exponentially
C) No current can flow
D) Zener breakdown occurs

Correct Answer: Option B | *Forward bias opposes built-in barrier, shrinking depletion layer and enabling exponential majority carrier diffusion.*

Q71. When a PN junction is reverse biased (P connected to negative, N to positive):

A) Depletion layer widens, barrier height increases (V0 + VR), and only a tiny minority saturation current I0 flows
B) Large diffusion current flows
C) Barrier potential drops to zero
D) Diode acts as a closed switch

Correct Answer: Option A | *Reverse bias aids built-in field, expanding depletion width and restricting current to reverse saturation current I0.*

Q72. The reverse saturation current I0 of a PN junction diode approximately:

A) Halves for every 10 °C rise in temperature
B) Doubles for every 10 °C rise in temperature
C) Remains strictly independent of temperature
D) Increases linearly with applied reverse voltage

Correct Answer: Option B | *I0 is governed by thermally generated minority carriers and roughly doubles every 10 °C increase in temperature.*

Q73. Zener breakdown occurs in PN junctions that are:

A) Lightly doped with wide depletion region under high reverse voltage (> 6 V)
B) Heavily doped with extremely narrow depletion region (< 10 nm) under moderate reverse voltage (< 6 V)
C) Forward biased at 0.7 V
D) Completely undoped

Correct Answer: Option B | *Zener breakdown occurs in heavily doped junctions where intense electric field (> 10■ V/cm) directly ruptures covalent bonds via quantum tunneling.*

Q74. Avalanche breakdown in PN junctions is caused by:

A) Direct electric field emission of valence electrons
B) Impact ionization where accelerated minority carriers collide with lattice atoms, generating secondary electron-hole pairs
C) High forward current
D) Optical excitation

Correct Answer: Option B | *Avalanche breakdown occurs in lightly doped diodes under higher reverse voltage via cumulative impact ionization.*

Q75. The temperature coefficient of breakdown voltage is _____ for Zener breakdown and _____ for Avalanche breakdown.

A) Positive; Negative
B) Negative; Positive
C) Zero; Zero
D) Positive; Positive

Correct Answer: Option B | *Zener breakdown voltage decreases with temperature (negative temp coeff), while Avalanche breakdown voltage increases with temperature (positive temp coeff).*

Q76. A Zener diode is operated in _____ bias to function as a _____.

A) Forward; Rectifier
B) Reverse breakdown region; Constant voltage regulator
C) Forward; Variable capacitor
D) Reverse; Solar energy collector

Correct Answer: Option B | *In reverse breakdown, Zener diode maintains an almost perfectly constant voltage across its terminals despite current variations.*

Q77. In a Bipolar Junction Transistor (BJT), the correct relationship among terminal currents is:

A) IC = IE + IB
B) IE = IB + IC
C) IB = IE + IC
D) IE = IC - IB

Correct Answer: Option B | *By KCL applied to BJT package, total emitter current is sum of base and collector currents: IE = IB + IC.*

Q78. The doping concentrations of the three BJT regions in descending order are:

A) Collector > Base > Emitter
B) Emitter > Collector > Base
C) Base > Emitter > Collector
D) Collector > Emitter > Base

Correct Answer: Option B | *Emitter is heavily doped (to inject carriers), Collector is moderately doped, and Base is very lightly doped (and ultra-thin).*

Q79. The physical sizes (areas) of the three BJT regions in descending order are:

A) Emitter > Base > Collector
B) Collector > Emitter > Base
C) Base > Collector > Emitter
D) Emitter > Collector > Base

Correct Answer: Option B | *Collector has the largest physical area to dissipate heat generated during collection of charge carriers.*

Q80. For a BJT to operate in the Active Region (linear amplifier mode):

A) Emitter-Base (EB) junction is Forward biased; Collector-Base (CB) junction is Reverse biased
B) EB junction is Reverse biased; CB junction is Forward biased
C) Both EB and CB junctions are Forward biased
D) Both EB and CB junctions are Reverse biased

Correct Answer: Option A | *Active mode requires forward-biased EB junction (injection) and reverse-biased CB junction (collection).*

Q81. For a BJT to operate as a closed switch (ON state in saturation):

A) Both EB and CB junctions are Forward biased
B) Both EB and CB junctions are Reverse biased
C) EB forward biased, CB reverse biased
D) EB reverse biased, CB forward biased

Correct Answer: Option A | *In Saturation region, both junctions are forward biased, giving minimal VCE,sat (~0.2 V) and maximum current.*

Q82. For a BJT to operate as an open switch (OFF state in cut-off):

A) Both EB and CB junctions are Reverse biased
B) Both EB and CB junctions are Forward biased
C) EB forward biased, CB reverse biased
D) Base is shorted to collector

Correct Answer: Option A | *In Cut-off region, both junctions are reverse biased, resulting in zero terminal currents (open switch).*

Q83. The Common Base DC current gain (α) and Common Emitter DC current gain (β) are defined as:

A) $\alpha = IC / IE$ and $\beta = IC / IB$
B) $\alpha = IB / IC$ and $\beta = IE / IC$
C) $\alpha = IE / IC$ and $\beta = IC / IB$
D) $\alpha = IC / IB$ and $\beta = IC / IE$

Correct Answer: Option A | *$\alpha = IC / IE$ (typically 0.95 to 0.998) and $\beta = IC / IB$ (typically 20 to 500).*

Q84. The mathematical relationship between β and α is:

A) $\beta = \alpha / (1 + \alpha)$
B) $\beta = \alpha / (1 - \alpha)$
C) $\beta = (1 - \alpha) / \alpha$
D) $\beta = 1 / (1 - \alpha)$

Correct Answer: Option B | *Since $IE = IB + IC$, dividing by IC gives $1/\alpha = 1/\beta + 1$, which rearranges to $\beta = \alpha / (1 - \alpha)$.*

Q85. If a transistor has $\alpha = 0.99$, its common emitter current gain β is:

A) 49
B) 99
C) 100
D) 0.01

Correct Answer: Option B | *$\beta = \alpha / (1 - \alpha) = 0.99 / (1 - 0.99) = 0.99 / 0.01 = 99$.*

Q86. A CMOS inverter is constructed using:

A) Two NMOS transistors in series
B) One PMOS transistor as pull-up network and one NMOS transistor as pull-down network
C) Two PMOS transistors in parallel
D) One BJT and one diode

Correct Answer: Option B | *CMOS inverter pairs a PMOS pull-up (connected to VDD) and an NMOS pull-down (connected to GND).*

Q87. In a CMOS inverter, when the input Vin is at Logic 1 (VDD):

A) PMOS is ON, NMOS is OFF, Vout = VDD
B) PMOS is OFF, NMOS is ON, Vout = 0 V (Logic 0)
C) Both transistors are ON
D) Both transistors are OFF

Correct Answer: Option B | *High input turns NMOS ON and PMOS OFF, pulling output cleanly down to 0 V (GND).*

Q88. The primary advantage of CMOS technology in computer logic design is:

A) High static power dissipation
B) Near-zero static power dissipation because PMOS and NMOS are never ON simultaneously in steady state
C) Requirement of negative supply voltage
D) High output impedance

Correct Answer: Option B | *In steady state (0 or 1), one transistor is always OFF, eliminating direct path from VDD to GND and yielding zero static power.*

Q89. A solar cell converts solar radiation directly into electrical energy based on the:

A) Thermionic emission
B) Photovoltaic effect across a PN junction
C) Seebeck effect
D) Hall effect

Correct Answer: Option B | *Solar cells operate via photovoltaic effect: light generates electron-hole pairs, separated by the junction field to create photovoltage.*

Q90. In the I-V characteristic curve of a solar cell, the cell operates in the:

A) First quadrant (power absorbing)
B) Second quadrant
C) Fourth quadrant (power delivering, V > 0 and I < 0)
D) Third quadrant

Correct Answer: Option C | *Solar cells deliver power to an external load, operating in the 4th quadrant of the I-V plane.*

Q91. The Fill Factor (FF) of a solar cell is defined as:

A) FF = (Voc · Isc) / (Vmp · Imp)
B) FF = (Vmp · Imp) / (Voc · Isc) = Pmax / (Voc · Isc)
C) FF = Voc / Isc
D) FF = Pout / Pin

Correct Answer: Option B | *FF is the ratio of maximum deliverable power (Vmp · Imp) to the product of open-circuit voltage Voc and short-circuit current Isc.*

Q92. A solar cell has Voc = 0.6 V, Isc = 3.0 A, and maximum power point values Vmp = 0.5 V, Imp = 2.7 A. The Fill Factor is:

A) 0.50
B) 0.75 (75%)
C) 0.90
D) 1.25

Correct Answer: Option B | *FF = (Vmp · Imp) / (Voc · Isc) = (0.5 * 2.7) / (0.6 * 3.0) = 1.35 / 1.80 = 0.75.*

Q93. Light propagates through the core of an optical fiber via:

A) Total Internal Reflection (TIR) at the core-cladding boundary
B) Continuous refraction through cladding
C) Electron conduction
D) Magnetic induction

Correct Answer: Option A | *Light is trapped and guided in optical fiber core through continuous Total Internal Reflection.*

Q94. For Total Internal Reflection to occur in an optical fiber, the refractive indices must satisfy:

A) Core refractive index n1 < Cladding refractive index n2
B) Core refractive index n1 > Cladding refractive index n2
C) n1 = n2
D) Cladding must be metallic

Correct Answer: Option B | *TIR requires light to travel from denser core (n1) toward rarer cladding (n2), so n1 > n2.*

Q95. The critical angle θc at the core-cladding interface is given by:

A) θc = sin■¹(n1 / n2)
B) θc = sin■¹(n2 / n1)
C) θc = cos■¹(n2 / n1)
D) θc = tan■¹(n1 / n2)

Correct Answer: Option B | *By Snell's Law, critical angle θc = sin■¹(n2 / n1). Light incident at angle > θc undergoes total internal reflection.*

Q96. Numerical Aperture (NA) of an optical fiber measures its light gathering capacity and is defined in terms of acceptance angle θa as:

A) NA = sin(θa) = √(n1² - n2²)
B) NA = cos(θa) = √(n1² + n2²)
C) NA = tan(θa) = n1 / n2
D) NA = n1 - n2

Correct Answer: Option A | *NA = sin(θa) = √(n1² - n2²) ≈ n1 √(2Δ), where Δ is fractional index difference.*

Q97. An optical fiber has core refractive index n1 = 1.50 and cladding index n2 = 1.48. The Numerical Aperture (NA) is:

A) 0.020
B) 0.244
C) 0.500
D) 0.987

Correct Answer: Option B | *NA = √(1.50² - 1.48²) = √(2.25 - 2.1904) = √0.0596 ≈ 0.2441.*

Q98. The condition for single-mode fiber (SMF) propagation in a step-index optical fiber is that the V-number must satisfy:

A) V > 100
B) V ≤ 2.405
C) V = 0
D) V ≥ 5.5

Correct Answer: Option B | *A step-index fiber operates strictly in single mode (HE11) when normalized frequency V ≤ 2.405.*

Q99. The total number of guided propagation modes M in a multimode step-index optical fiber with large V-number is approximately:

A) M ≈ V
B) M ≈ V² / 2
C) M ≈ V² / 4
D) M ≈ 2 V²

Correct Answer: Option B | *For step-index multimode fiber, M ≈ V² / 2. (For parabolic graded-index fiber, M ≈ V² / 4).*

Q100. A CPU architecture is fundamentally optimized for _____, whereas a GPU architecture is optimized for _____.

A) High data throughput; Ultra-low instruction latency
B) Ultra-low instruction latency with deep caches; Massive parallel data throughput with thousands of ALUs
C) Graphics processing only; Serial pointer arithmetic only
D) Zero memory bandwidth; High clock jitter

Correct Answer: Option B | *CPUs optimize single-thread latency with large caches and branch prediction; GPUs optimize parallel throughput with thousands of SIMD cores.*

Q101. In an electric circuit, electric power P dissipated in a resistor R carrying current I with voltage V is:

A) P = V · I = I² R = V² / R
B) P = V / I = I / R
C) P = V² · R
D) P = I / V²

Correct Answer: Option A | *Joule's law: P = V I = I² R = V² / R.*

Q102. An electric bulb rated 100 W, 200 V has an internal operating resistance of:

A) 2 Ω
B) 200 Ω
C) 400 Ω
D) 20,000 Ω

Correct Answer: Option C | *R = V² / P = (200)² / 100 = 40,000 / 100 = 400 Ω.*

Q103. If three resistors of 6 Ω, 3 Ω, and 2 Ω are connected in parallel, the equivalent resistance is:

A) 11 Ω
B) 1 Ω
C) 0.5 Ω
D) 3 Ω

Correct Answer: Option B | *1/Req = 1/6 + 1/3 + 1/2 = 1/6 + 2/6 + 3/6 = 6/6 = 1 Ω■¹ => Req = 1 Ω.*

Q104. The relationship between collector current IC, base current IB, and reverse saturation current ICBO in common emitter mode is:

A) IC = β IB + (1 + β) ICBO
B) IC = β IB + ICBO
C) IC = α IB + ICBO
D) IC = β IE

Correct Answer: Option A | *IC = α IE + ICBO = α (IB + IC) + ICBO => IC = (α/(1-α)) IB + (1/(1-α)) ICBO = β IB + (1 + β) ICBO.*

Q105. In a BJT amplifier, the Common Collector (Emitter Follower) configuration provides:

A) High voltage gain and low current gain
B) Voltage gain ≈ 1, high current gain, high input impedance, and low output impedance (ideal buffer)
C) High voltage gain and high output impedance
D) Zero power gain

Correct Answer: Option B | *Common Collector acts as a voltage buffer (Av ≈ 1) with high Zin and low Zout, perfect for impedance matching.*

Q106. The Common Emitter (CE) configuration of BJT provides a phase shift between input and output voltage of:

A) 0° (in-phase)
B) 90°
C) 180° (out-of-phase)
D) 270°

Correct Answer: Option C | *CE configuration introduces a 180° phase inversion between input signal and amplified output voltage.*

Q107. Fractional refractive index difference Δ in an optical fiber is defined as:

A) Δ = (n1 - n2) / n1
B) Δ = (n1 + n2) / n1
C) Δ = (n1 - n2) / n2
D) Δ = n1 · n2

Correct Answer: Option A | *Fractional index difference Δ = (n1 - n2) / n1, representing the relative step between core and cladding.*

Q108. In a graded-index (GRIN) optical fiber, the refractive index of the core:

A) Is strictly constant throughout the core
B) Decreases parabolically from center axis toward the cladding
C) Increases linearly toward the cladding
D) Is lower than the cladding index

Correct Answer: Option B | *GRIN core index decreases parabolically n(r) = n1 √(1 - 2Δ(r/a)²), equalizing transit times of different optical rays.*

Q109. Intermodal dispersion is completely absent in which type of optical fiber?

A) Multimode Step-Index Fiber
B) Multimode Graded-Index Fiber
C) Single-Mode Fiber (SMF)
D) Plastic Optical Fiber

Correct Answer: Option C | *Single-mode fiber supports only one fundamental propagation mode, inherently eliminating intermodal dispersion.*

Q110. A step-index fiber has core radius a = 4 μm, NA = 0.15, operating at wavelength λ = 1.3 μm. The V-number is:

A) 2.90
B) 1.50
C) 5.80
D) 0.85

Correct Answer: Option A | *V = (2π a / λ) NA = (2 * 3.1416 * 4 / 1.3) * 0.15 = 19.333 * 0.15 ≈ 2.90.*

Q111. The power conversion efficiency (η) of a solar cell is given by:

A) η = (Voc · Isc · FF / Pin) × 100%
B) η = (Pin / Pmax) × 100%
C) η = (Voc / Isc) × 100%
D) η = (FF / Pin) × 100%

Correct Answer: Option A | *Efficiency η = (Pmax / Pin) × 100% = (Voc · Isc · FF / Pin) × 100%.*

Q112. A solar panel of area 1.5 m² is exposed to solar irradiance of 1000 W/m². If it delivers 225 W electrical power, its efficiency is:

A) 10%
B) 15%
C) 22.5%
D) 30%

Correct Answer: Option B | *Pin = 1000 W/m² * 1.5 m² = 1500 W. Efficiency* η *= (225 / 1500) × 100% = 15%.*

Q113. In a CE transistor amplifier, if base current changes by 20 μA and collector current changes by 2 mA, the AC current gain βac is:

A) 10
B) 100
C) 50
D) 200

Correct Answer: Option B | *βac = ΔIC / ΔIB = (2 × 10■³ A) / (20 × 10■■ A) = 100.*

Q114. The relationship between Common Collector current gain γ and Common Emitter current gain β is:

A) γ = β
B) γ = 1 + β
C) γ = β - 1
D) γ = β / (1 + β)

Correct Answer: Option B | *γ = IE / IB = (IB + IC) / IB = 1 +* β.

Q115. In an NPN transistor, the majority charge carriers in the emitter and collector are:

A) Holes
B) Electrons
C) Positive ions
D) Photons

Correct Answer: Option B | *In NPN BJT, Emitter (N) and Collector (N) have electrons as majority carriers, while Base (P) has holes.*

Q116. In a PNP transistor, the majority charge carriers injected from emitter into base are:

A) Holes
B) Electrons
C) Negative ions
D) Neutrons

Correct Answer: Option A | *In PNP BJT, Emitter is P-type, so it injects holes into the N-type base region.*

Q117. A diode circuit has an AC dynamic resistance rd defined as:

A) rd = V / I
B) rd = ΔV / ΔI = η VT / ID
C) rd = ID / VT
D) rd = 1 / (V · I)

Correct Answer: Option B | *Dynamic resistance is the slope reciprocal rd = dV/dI =* η *VT / ID ≈ 26 mV / ID at room temperature.*

Q118. For a Silicon diode carrying forward current ID = 26 mA at 300 K (taking ideality factor η = 1), dynamic resistance rd is:

A) 1 Ω
B) 26 Ω
C) 0.1 Ω
D) 10 Ω

Correct Answer: Option A | *rd = VT / ID = 26 mV / 26 mA = 1* Ω.

Q119. The ideal diode equation (Shockley equation) for current I as a function of voltage V is:

A) I = I0 [exp(qV / η kB T) - 1]
B) I = I0 [exp(η kB T / qV) + 1]
C) I = I0 (qV / kB T)
D) I = V / R

Correct Answer: Option A | *Shockley diode equation is I = I0 [exp(V /* η *VT) - 1].*

Q120. When a reverse bias VR >> VT is applied across a PN diode, the diode current approaches:

A) 0 A
B) -I0 (reverse saturation current)
C) +I0
D) -∞

Correct Answer: Option B | *When V is negative and |V| >> VT, exp(V/VT) → 0, so I = I0 (0 - 1) = -I0.*

Q121. In a bridge rectifier circuit with sinusoidal AC input, the number of diodes conducting simultaneously in each half cycle is:

A) 1
B) 2
C) 4
D) 0

Correct Answer: Option B | *In a bridge rectifier, two diagonally opposite diodes conduct simultaneously during each half cycle.*

Q122. The ripple factor of an ideal full-wave rectifier without filter is approximately:

A) 1.21
B) 0.482
C) 0.10
D) 0.0

Correct Answer: Option B | *Full-wave rectifier ripple factor is $\gamma = \sqrt{((I_{rms}/I_{dc})^2 - 1)} \approx 0.482$ (compared to 1.21 for half-wave).*

Q123. The maximum theoretical rectification efficiency of a full-wave rectifier is:

A) 40.6%
B) 81.2%
C) 50.0%
D) 100%

Correct Answer: Option B | *Full-wave rectifier maximum theoretical efficiency is 81.2% (twice that of a half-wave rectifier, 40.6%).*

Q124. In a BJT, the Early Effect (base-width modulation) is caused by:

A) Variation of reverse-bias voltage across the collector-base junction altering the effective base width
B) High temperature melting the base
C) Heavy doping of emitter
D) Forward biasing collector

Correct Answer: Option A | *Higher reverse collector bias widens the CB depletion layer into the base, reducing the effective neutral base width.*

Q125. Which BJT configuration is most widely used for voltage and power amplification?

A) Common Base (CB)
B) Common Emitter (CE)
C) Common Collector (CC)
D) Inverted Collector

Correct Answer: Option B | *Common Emitter provides both high voltage gain and high current gain, yielding the highest power gain.*

Q126. In a CMOS circuit, dynamic power dissipation during switching is given by:

A) $P_{dyn} = C_{load} \cdot V_{DD}^2 \cdot f$
B) $P_{dyn} = V_{DD} \cdot I_{static}$
C) $P_{dyn} = C_{load} \cdot V_{DD} / f$
D) $P_{dyn} = V_{DD}^2 / R$

Correct Answer: Option A | *Dynamic switching power is $P_{dyn} = \alpha \cdot C_{load} \cdot V_{DD}^2 \cdot f$, where f is switching clock frequency.*

Q127. SIMD (Single Instruction, Multiple Data) execution model is characteristic of:

A) Traditional single-core CPU control logic
B) Modern GPU compute architectures for vector/matrix processing
C) Hard disk drives
D) Static RAM chips

Correct Answer: Option B | *GPUs use SIMD/SIMT architectures where a single instruction stream operates on thousands of data points simultaneously.*

Q128. Which component in a CPU architecture is responsible for predicting conditional jumps to avoid pipeline stalls?

A) Arithmetic Logic Unit (ALU)
B) Branch Predictor and Speculative Execution Unit
C) Level 3 Cache
D) Memory Management Unit (MMU)

Correct Answer: Option B | *Branch prediction speculatively predicts execution paths to keep instruction pipelines full and reduce latency.*

Q129. An optical fiber has NA = 0.20 and core index n1 = 1.46. The acceptance angle θa in air (n0 = 1.0) is:

A) 11.54°
B) 23.08°
C) 45.0°
D) 6.0°

Correct Answer: Option A | *θa = sin■¹(NA) = sin■¹(0.20) ≈ 11.54°.*

Q130. If the fractional index difference of a fiber is Δ = 0.01 and n1 = 1.5, the cladding index n2 is:

A) 1.485
B) 1.490
C) 1.400
D) 1.515

Correct Answer: Option A | *Δ = (n1 - n2) / n1 => 0.01 = (1.5 - n2) / 1.5 => 0.015 = 1.5 - n2 => n2 = 1.485.*

Q131. A Zener diode voltage regulator circuit has unregulated input Vin = 15 V, regulated Zener voltage VZ = 9 V, and series resistor Rs = 100 Ω. The total current through Rs is:

A) 60 mA
B) 90 mA
C) 150 mA
D) 6 mA

Correct Answer: Option A | *Current through Rs is Is = (Vin - VZ) / Rs = (15 - 9) / 100 = 6 / 100 = 0.06 A = 60 mA.*

Q132. In an optical fiber, attenuation (loss) is primarily caused by:

A) Rayleigh scattering and material absorption (OH■ absorption bands)
B) Cosmic rays only
C) Light travelling too fast
D) Magnetic interference from Earth

Correct Answer: Option A | *Fiber losses stem from intrinsic Rayleigh scattering (∝ 1/λ■) and absorption bands (such as water vapor OH■ ions).*

Q133. The standard telecommunication optical window with the lowest fiber attenuation (~0.2 dB/km) is at wavelength:

A) 850 nm
B) 1310 nm
C) 1550 nm
D) 632 nm

Correct Answer: Option C | *Silica fiber has its global attenuation minimum (~0.2 dB/km) in the 1550 nm (C-band) window.*

Q134. A BJT has collector current IC = 4.9 mA and base current IB = 100 μA. The emitter current IE is:

A) 4.8 mA
B) 5.0 mA
C) 5.9 mA
D) 4.9 mA

Correct Answer: Option B | *IE = IC + IB = 4.9 mA + 0.1 mA = 5.0 mA.*

Q135. For the transistor in Q134 (IC = 4.9 mA, IE = 5.0 mA, IB = 0.1 mA), the current gains α and β are:

A) α = 0.98, β = 49
B) α = 0.49, β = 98
C) α = 0.99, β = 100
D) α = 0.95, β = 19

Correct Answer: Option A | *α = IC / IE = 4.9 / 5.0 = 0.98; β = IC / IB = 4.9 / 0.1 = 49.*

■ UNIT 3: DIGITAL ELECTRONICS, LOGIC & K-MAPS

Q136. The binary equivalent of the decimal number (25)■■ is:

A) (11001)■
B) (11010)■
C) (10011)■
D) (11101)■

Correct Answer: Option A | *25 = 16 + 8 + 1 = 11001 in binary (1·16 + 1·8 + 0·4 + 0·2 + 1·1).*

Q137. The hexadecimal equivalent of the binary number (110110111100)■ is:

A) (DBC)■■
B) (EBC)■■
C) (DBE)■■
D) (ABC)■■

Correct Answer: Option A | *Group 4 bits from right: 1101 = D (13), 1011 = B (11), 1100 = C (12) => (DBC)■■.*

Q138. The octal equivalent of the binary number (101110010)■ is:

A) (562)■
B) (572)■
C) (275)■
D) (652)■

Correct Answer: Option A | *Group 3 bits from right: 101 = 5, 110 = 6, 010 = 2 => (562)■.*

Q139. The 1's complement of the binary number 10110100 is obtained by:

A) Adding 1 to the number
B) Inverting each bit (0→1, 1→0), yielding 01001011
C) Inverting only the MSB
D) Shifting bits to left

Correct Answer: Option B | *1's complement bit inversion: 10110100 becomes 01001011.*

Q140. The 2's complement of the 8-bit binary number 00101100 is:

A) 11010011
B) 11010100
C) 11010001
D) 00101101

Correct Answer: Option B | *1's complement = 11010011. Adding 1 gives 11010011 + 1 = 11010100. (Or keep bits up to first 1 from right, invert rest).*

Q141. In an 8-bit 2's complement signed number system, the decimal range of representable numbers is:

A) -127 to +127
B) -128 to +127
C) 0 to 255
D) -256 to +255

Correct Answer: Option B | *For n-bit 2's complement, range is -2^(n-1) to +(2^(n-1) - 1). For n=8, -128 to +127.*

Q142. A major advantage of 2's complement representation over 1's complement is that 2's complement:

A) Has two distinct representations for zero (+0 and -0)
B) Has a unique representation for zero (00000000) and simplified subtraction circuitry
C) Requires more hardware
D) Cannot represent negative numbers

Correct Answer: Option B | *2's complement eliminates negative zero (-0) and allows subtraction to be performed directly with adder hardware by discarding the final end-carry.*

Q143. When subtracting binary numbers using 1's complement, if an end-around carry is generated, it must be:

A) Discarded
B) Added to the Least Significant Bit (LSB)
C) Subtracted from MSB
D) Multiplied by 2

Correct Answer: Option B | *In 1's complement arithmetic, an end carry indicates a positive result and must be added back to the LSB (end-around carry).*

Q144. When subtracting binary numbers using 2's complement, if a final end carry is generated:

A) The result is positive and the end carry is discarded
B) The end carry is added to the LSB
C) The result is negative and must be complemented
D) An arithmetic overflow has definitely occurred

Correct Answer: Option A | *In 2's complement, an end carry signifies the result is positive and the carry is simply discarded.*

Q145. The BCD (Binary Coded Decimal / 8421 code) representation of decimal number 87 is:

A) 1010111
B) 1000 0111
C) 0111 1000
D) 1000 0101

Correct Answer: Option B | *Encode each decimal digit into 4 bits: 8 = 1000, 7 = 0111 => 1000 0111.*

Q146. Which of the following 4-bit combinations is an INVALID BCD code?

A) 0101
B) 1001
C) 1100
D) 0000

Correct Answer: Option C | *Valid BCD covers 0000 (0) to 1001 (9). 1010 (10) through 1111 (15) are invalid BCD states.*

Q147. The Excess-3 code for decimal number 5 is:

A) 0101
B) 1000
C) 0011
D) 1010

Correct Answer: Option B | *Excess-3 = Decimal + 3 = 5 + 3 = 8 => (1000)■ in 4-bit binary.*

Q148. Excess-3 code is known as a 'self-complementing code' because:

A) Its 1's complement produces the 9's complement of the corresponding decimal digit
B) Its 2's complement is 0
C) It uses only NAND gates
D) It has unit Hamming distance

Correct Answer: Option A | *Inverting all bits of an Excess-3 code word gives the Excess-3 code of the 9's complement of that decimal digit.*

Q149. Gray Code is called a 'unit distance code' because:

A) Every code word has exactly one '1'
B) Consecutive code words differ in only one bit position
C) It represents numbers in distance units
D) It has zero parity

Correct Answer: Option B | *Gray code ensures only one bit flips between any two adjacent numbers, preventing spurious glitches/hazards in encoders and K-maps.*

Q150. The Gray code equivalent of binary number 1011 is:

A) 1110
B) 1101
C) 0110
D) 1001

Correct Answer: Option A | *G3 = B3 = 1; G2 = B3 ⊕ B2 = 1 ⊕ 0 = 1; G1 = B2 ⊕ B1 = 0 ⊕ 1 = 1; G0 = B1 ⊕ B0 = 1 ⊕ 1 = 0 => 1110.*

Q151. The binary equivalent of Gray code 1101 is:

A) 1001
B) 1000
C) 1010
D) 1111

Correct Answer: Option A | *B3 = G3 = 1; B2 = B3 ⊕ G2 = 1 ⊕ 1 = 0; B1 = B2 ⊕ G1 = 0 ⊕ 0 = 0; B0 = B1 ⊕ G0 = 0 ⊕ 1 = 1 => 1001.*

Q152. Which of the following pairs of logic gates are called Universal Logic Gates?

A) AND and OR
B) NAND and NOR
C) XOR and XNOR
D) NOT and AND

Correct Answer: Option B | *NAND and NOR can independently implement any arbitrary Boolean function (AND, OR, NOT, XOR, XNOR).*

Q153. The minimum number of 2-input NAND gates required to implement a 2-input XOR gate is:

A) 3
B) 4
C) 5
D) 2

Correct Answer: Option B | *XOR gate requires exactly 4 NAND gates. (XNOR requires 5 NAND gates).*

Q154. The minimum number of 2-input NOR gates required to implement a 2-input XOR gate is:

A) 4
B) 5
C) 3
D) 2

Correct Answer: Option B | *XOR gate requires 5 NOR gates. (XNOR requires 4 NOR gates).*

Q155. The minimum number of 2-input NAND gates required to implement a 2-input AND gate is:

A) 1
B) 2
C) 3
D) 4

Correct Answer: Option B | *NAND followed by NAND-as-inverter gives (A·B)'' = A·B, requiring 2 NAND gates.*

Q156. The minimum number of 2-input NAND gates required to implement a 2-input OR gate is:

A) 2
B) 3
C) 4
D) 1

Correct Answer: Option B | *By De Morgan's, A + B = (A' · B')'. Invert A (1 NAND), invert B (1 NAND), then NAND together (1 NAND) = 3 NAND gates.*

Q157. According to De Morgan's First Law, (A + B)' is equivalent to:

A) A' + B'
B) A' · B'
C) A · B
D) (A · B)'

Correct Answer: Option B | *De Morgan's Theorem 1: Complement of a sum equals product of complements: (A + B)' = A' · B'.*

Q158. According to De Morgan's Second Law, (A · B)' is equivalent to:

A) A' · B'
B) A' + B'
C) A + B
D) (A + B)'

Correct Answer: Option B | *De Morgan's Theorem 2: Complement of a product equals sum of complements: (A · B)' = A' + B'.*

Q159. The Boolean expression A + A'B simplifies to:

A) A
B) A + B
C) A · B
D) B

Correct Answer: Option B | *By Distributive Law: A + A'B = (A + A')(A + B) = 1 · (A + B) = A + B.*

Q160. The Boolean expression A(A' + B) simplifies to:

A) A + B
B) A · B
C) A'
D) 0

Correct Answer: Option B | *A · A' + A · B = 0 + A · B = A · B.*

Q161. According to the Consensus Theorem, the expression AB + A'C + BC simplifies to:

A) AB + A'C
B) BC
C) AB + BC
D) A'C + BC

Correct Answer: Option A | *Consensus theorem: BC is the redundant consensus term of AB and A'C, so AB + A'C + BC = AB + A'C.*

Q162. The dual of the Boolean expression A + B·C = (A + B)(A + C) is obtained by:

A) Replacing + with ·, · with +, and swapping 0 and 1, yielding A · (B + C) = (A · B) + (A · C)
B) Complementing every literal
C) Inverting only the LHS
D) Multiplying by 2

Correct Answer: Option A | *Duality principle: Swap AND (·) and OR (+), and swap 0 and 1 without complementing literals.*

Q163. In a Karnaugh Map (K-Map), adjacent cells are arranged according to:

A) Natural binary sequence (00, 01, 10, 11)
B) Gray code sequence (00, 01, 11, 10)
C) BCD code
D) Excess-3 code

Correct Answer: Option B | *K-Map columns and rows use Gray code (00, 01, 11, 10) so adjacent physical cells differ by exactly one bit.*

Q164. A 4-variable K-Map contains a total of _____ cells.

A) 4
B) 8
C) 16
D) 32

Correct Answer: Option C | *Number of cells for n variables is 2■. For n=4, 2■ = 16 cells.*

Q165. In a K-Map, valid cell groupings must contain a number of 1s equal to:

A) Any integer (1, 2, 3, 4, 5...)
B) A power of 2 (1, 2, 4, 8, 16...)
C) Multiples of 3
D) Odd numbers only

Correct Answer: Option B | *Valid groups must be powers of 2 (2^k) in rectangular or square shapes.*

Q166. In a 4-variable K-map for F(A,B,C,D), grouping the 4 corner cells (m0, m2, m8, m10) simplifies to:

A) A'C'
B) B'D'
C) BD
D) A'D'

Correct Answer: Option B | *m0(0000), m2(0010), m8(1000), m10(1010). Row A varies (0,1), B is constant 0 (B'). Column C varies (0,1), D is constant 0 (D') => B'D'.*

Q167. A 'Don't Care' condition (d or X) in a K-Map:

A) Must always be included in every group
B) Must never be grouped with 1s
C) Can be treated as 1 if it helps form a larger group, or treated as 0 otherwise
D) Inverts the output to 0

Correct Answer: Option C | *Don't Care terms provide flexibility: include as 1 only if they enlarge a group to minimize literals; ignore as 0 otherwise.*

Q168. An Essential Prime Implicant (EPI) is a prime implicant that:

A) Covers at least one minterm '1' that is not covered by any other prime implicant
B) Covers all minterms
C) Contains don't care conditions only
D) Can be safely deleted without affecting logic

Correct Answer: Option A | *EPI must be included in the minimal SOP expression because it uniquely covers at least one 1-cell.*

Q169. The canonical Sum of Products (SOP) form of a logic function represents the logic using:

A) Maxterms (Π M)
B) Minterms (Σ m)
C) NOR gates only
D) XOR gates only

Correct Answer: Option B | *Canonical SOP is a sum (OR) of fundamental product terms (minterms Σ m).*

Q170. The canonical Product of Sums (POS) form of a logic function represents the logic using:

A) Minterms (Σ m)
B) Maxterms (Π M)
C) NAND gates only
D) XNOR gates only

Correct Answer: Option B | *Canonical POS is a product (AND) of fundamental sum terms (maxterms Π M).*

Q171. For a 3-variable function F(A,B,C), the minterm m5 corresponds to:

A) A' B C'
B) A B' C
C) A B C'
D) A' B' C

Correct Answer: Option B | *5 in binary is 101 => A=1, B=0, C=1 => product term A B' C.*

Q172. For a 3-variable function F(A,B,C), the maxterm M5 corresponds to:

A) A' + B + C'
B) A + B' + C
C) A' + B' + C
D) A + B + C'

Correct Answer: Option A | *For maxterms, 0 is uncomplemented and 1 is complemented. 5 = 101 => A' + B + C'.*

Q173. The output of a 2-input XOR gate is Logic 1 if and only if:

A) Both inputs are 1
B) Both inputs are 0
C) The two inputs are different (one is 0, other is 1)
D) Both inputs are identical

Correct Answer: Option C | *XOR (exclusive OR) outputs 1 when inputs are unequal (odd parity detector: 0⊕1=1, 1⊕0=1).*

Q174. The output of a 2-input XNOR gate (Equivalence Gate) is Logic 1 when:

A) The two inputs are identical (both 0 or both 1)
B) Inputs are different
C) Input A is 1 only
D) Input B is 1 only

Correct Answer: Option A | *XNOR outputs 1 when inputs are equal (0■0=1, 1■1=1), acting as a coincidence detector.*

Q175. The logic expression (A ⊕ B) ⊕ B simplifies to:

A) A
B) B
C) 1
D) 0

Correct Answer: Option A | *By associativity and property B ⊕ B = 0: (A ⊕ B) ⊕ B = A ⊕ (B ⊕ B) = A ⊕ 0 = A.*

Q176. The Boolean identity A ⊕ 1 is equal to:

A) A
B) A' (NOT A)
C) 1
D) 0

Correct Answer: Option B | *A ⊕ 1 = A·1' + A'·1 = 0 + A' = A'.*

Q177. The Boolean identity A ⊕ 0 is equal to:

A) A
B) A'
C) 0
D) 1

Correct Answer: Option A | *A ⊕ 0 = A·0' + A'·0 = A·1 + 0 = A.*

Q178. In a 3-variable K-map for F(A,B,C) = Σ m(0, 1, 2, 3), the minimal simplified expression is:

A) A'
B) B'
C) C'
D) A

Correct Answer: Option A | *m0, m1, m2, m3 occupy the entire top row where A=0 => A'.*

Q179. In a 3-variable K-map for F(A,B,C) = Σ m(1, 3, 5, 7), the minimal simplified expression is:

A) A
B) B
C) C
D) C'

Correct Answer: Option C | *All four minterms have C=1 across all combinations of A and B => C.*

Q180. In a 4-variable K-map, grouping an entire column of 4 cells eliminates _____ variables, leaving a product term with _____ variables.

A) 1; 3
B) 2; 2
C) 3; 1
D) 4; 0

Correct Answer: Option B | *A group of 4 cells (2^2) eliminates log2(4) = 2 variables, leaving 4 - 2 = 2 variables.*

Q181. In a 4-variable K-map, grouping an entire block of 8 cells eliminates _____ variables, leaving a single literal with _____ variable(s).

A) 3; 1
B) 2; 2
C) 1; 3
D) 4; 0

Correct Answer: Option A | *An octet (8 cells = 2^3) eliminates log2(8) = 3 variables, reducing to 4 - 3 = 1 variable.*

Q182. The 9's complement of decimal number 438 is:

A) 561
B) 562
C) 651
D) 560

Correct Answer: Option A | *Subtract each digit from 9: 999 - 438 = 561.*

Q183. The 10's complement of decimal number 438 is:

A) 561
B) 562
C) 560
D) 652

Correct Answer: Option B | *10's complement = 9's complement + 1 = 561 + 1 = 562.*

Q184. Convert binary (11010.11)■ to decimal:

A) 26.75
B) 26.50
C) 24.75
D) 26.25

Correct Answer: Option A | *16 + 8 + 0 + 2 + 0 = 26; Fractional part = $1 \cdot 2^{-1} + 1 \cdot 2^{-2}$ = 0.5 + 0.25 = 0.75 => 26.75.*

Q185. Convert hexadecimal (2F.4)■■ to decimal:

A) 47.25
B) 47.50
C) 31.25
D) 45.25

Correct Answer: Option A | *$2 \cdot 16^1 + 15 \cdot 16^{-1}$ = 32 + 15 = 47; Fractional part = 4 / 16 = 0.25 => 47.25.*

Q186. Add the binary numbers 1011 and 1101. The result is:

A) 11000
B) 10110
C) 11100
D) 10000

Correct Answer: Option A | *1011 (11) + 1101 (13) = 24 => (11000)■.*

Q187. In a digital system, an arithmetic overflow when adding two signed numbers occurs if:

A) Adding two positive numbers yields a negative result, or adding two negative numbers yields a positive result
B) An end carry is generated
C) Both numbers are zero
D) The result is positive

Correct Answer: Option A | *Overflow occurs in signed addition when adding two operands of the same sign produces a result of opposite sign (Cin ⊕ Cout at MSB = 1).*

Q188. The Boolean expression (A + B)(A + B') simplifies to:

A) A
B) B
C) A + B
D) 0

Correct Answer: Option A | *A·A + A·B' + B·A + B·B' = A + A·B' + A·B + 0 = A(1 + B' + B) = A·1 = A.*

Q189. The Boolean expression AB + AB' + A'B simplifies to:

A) A + B
B) A · B
C) A' + B'
D) 1

Correct Answer: Option A | *A(B + B') + A'B = A·1 + A'B = A + A'B = A + B.*

Q190. The minimum number of 2-input NOR gates required to implement a NOT gate (inverter) is:

A) 1
B) 2
C) 3
D) 4

Correct Answer: Option A | *Tie both inputs together: (A + A)' = A', requiring 1 NOR gate.*

Q191. The minimum number of 2-input NAND gates required to implement a NOT gate (inverter) is:

A) 1
B) 2
C) 3
D) 4

Correct Answer: Option A | *Tie both inputs together: (A · A)' = A', requiring 1 NAND gate.*

Q192. The minimum number of 2-input NOR gates required to implement an OR gate is:

A) 1
B) 2
C) 3
D) 4

Correct Answer: Option B | *NOR gate gives (A + B)'. Followed by a NOR inverter gives ((A + B)')' = A + B (2 NOR gates).*

Q193. The minimum number of 2-input NOR gates required to implement an AND gate is:

A) 2
B) 3
C) 4
D) 5

Correct Answer: Option B | *By De Morgan's, A · B = (A' + B')'. Invert A (1 NOR), invert B (1 NOR), and NOR them together (1 NOR) = 3 NOR gates.*

Q194. The Gray code sequence for 3-bit numbers starting from 0 to 7 is:

A) 000, 001, 011, 010, 110, 111, 101, 100
B) 000, 001, 010, 011, 100, 101, 110, 111
C) 000, 010, 001, 011, 100, 110, 101, 111
D) 111, 110, 101, 100, 011, 010, 001, 000

Correct Answer: Option A | *Standard 3-bit Gray code sequence is 000, 001, 011, 010, 110, 111, 101, 100.*

Q195. In a 4-variable K-map, if all 16 cells contain '1', the output simplifies to:

A) 1 (logic high)
B) 0
C) ABCD
D) A + B + C + D

Correct Answer: Option A | *A group of 16 cells (2■) in a 4-variable map eliminates all 4 variables, yielding constant 1.*

Q196. Simplify the Boolean function F(A,B,C) = Σ m(0, 2, 4, 6):

A) C'
B) B'
C) A'
D) C

Correct Answer: Option A | *m0(000), m2(010), m4(100), m6(110). A and B vary across all states, while C is constant 0 => C'.*

Q197. Simplify the Boolean function F(A,B,C) = Σ m(4, 5, 6, 7):

A) A
B) B
C) C
D) A'

Correct Answer: Option A | *m4, m5, m6, m7 span the entire bottom row where A=1 => A.*

Q198. In a Half Adder circuit, the Sum (S) and Carry (C) outputs for inputs A and B are:

A) S = A ⊕ B, C = A · B
B) S = A · B, C = A ⊕ B
C) S = A + B, C = A' · B
D) S = (A ⊕ B)', C = A + B

Correct Answer: Option A | *Half Adder performs modulo-2 addition: Sum = A ⊕ B and Carry = A · B.*

Q199. In a Full Adder circuit with inputs A, B, and Cin, the Sum (S) is given by:

A) S = A ⊕ B ⊕ Cin
B) S = A · B · Cin
C) S = A + B + Cin
D) S = (A ⊕ B) · Cin

Correct Answer: Option A | *Full Adder 3-input odd parity sum is S = A ⊕ B ⊕ Cin.*

Q200. The total number of minterms in a Boolean function of n variables is:

A) 2■
B) 2^(2■)
C) n²
D) 2n

Correct Answer: Option A | *An n-variable Boolean space contains exactly 2■ distinct minterms (and 2■ maxterms). The total number of distinct switching functions is 2^(2■).*